

Activité Pratique N°1 — Microservice Bank Account avec Spring Boot

Réalisé par : Rim Jabal Référence : <https://www.youtube.com/watch?v=2-qloZcvhAw>

Objectif

Développer un microservice de gestion de comptes bancaires avec Spring Boot, exposé de plusieurs façons : - une API REST classique (contrôleur + couche service + DTOs), - une API générée automatiquement avec **Spring Data REST** (avec projections), - une documentation **Swagger / OpenAPI**, - un service web **GraphQL**.

Technologies

- Java 17, Spring Boot 4.1.1, Maven
- Spring Web MVC, Spring Data JPA, Spring Data REST, Spring HATEOAS
- Base de données **H2** (en mémoire)
- Lombok
- Spring for GraphQL (+ GraphiQL)
- springdoc-openapi (Swagger UI)
- Postman

Architecture du projet

```
org.sid.bank_account_service
├─ dto/           BankAccountRequestDTO, BankAccountResponseDTO
├─ entities/      BankAccount, Customer, AccountProjection
├─ enums/         AccountType
├─ exceptions/    CustomDataFetcherExceptionResolver
├─ mappers/       BankAccountMapper
├─ repositories/  BankAccountRepository, CustomerRepository
├─ service/       AccountService, AccountServiceImpl
├─ web/           AccountRestController, BankAccountGraphQLController
├─ BankAccountServiceApplication
resources/
├─ application.properties
├─ graphql/schema.graphqls
```

1. Création du projet

Le projet a été généré avec **Spring Initializr** avec les dépendances : Spring Web, Spring Data JPA, H2 Database, Lombok et Spring for GraphQL. Les dépendances **Spring Data REST**, **Spring HATEOAS** et **springdoc-openapi** ont été ajoutées ensuite dans le `pom.xml`.

2. Entités JPA et repositories

- `BankAccount` : `id` (UUID), `createdAt`, `balance`, `currency`, `type` (enum `AccountType` : `SAVING_ACCOUNT` / `CURRENT_ACCOUNT`) et une relation `@ManyToOne` vers `Customer`.
- `Customer` : `id`, `name` et une relation `@OneToMany(mappedBy = "customer")` vers `BankAccount`.
L'annotation `@JsonProperty(access = WRITE_ONLY)` évite la récursivité infinie lors de la sérialisation JSON.
- `BankAccountRepository` et `CustomerRepository` héritent de `JpaRepository`.

Au démarrage, un `CommandLineRunner` crée 4 clients et 10 comptes par client.

Configuration (`application.properties`) :

```
server.port=8081
spring.datasource.url=jdbc:h2:mem:account-db
spring.h2.console.enabled=true
spring.graphql.graphiql.enabled=true
```

Console H2 (<http://localhost:8081/h2-console>) — les tables `BANK_ACCOUNT` et `CUSTOMER` sont bien créées :



3. Couche service, DTOs et mapper

- `BankAccountRequestDTO` : données envoyées par le client (`balance`, `currency`, `type`).
- `BankAccountResponseDTO` : données renvoyées (`id`, `createdAt`, `balance`, `currency`, `type`, `customer`).
- `BankAccountMapper` : conversion Entité ↔ DTO.
- `AccountService` / `AccountServiceImpl` (`@Service` `@Transactional`) : `accountList`, `accountById`, `addAccount`, `updateAccount`, `deleteAccount`.

4. API REST (`AccountRestController`)

Le contrôleur est mappé sur `/api` pour éviter les conflits avec les endpoints générés par Spring Data REST.

Méthode	Endpoint	Description
GET	/api/bankAccounts	Liste des comptes
GET	/api/bankAccounts/{id}	Compte par id
POST	/api/bankAccounts	Ajouter un compte
PUT	/api/bankAccounts/{id}	Modifier un compte
DELETE	/api/bankAccounts/{id}	Supprimer un compte

Liste des comptes (<http://localhost:8081/api/bankAccounts>) — chaque compte contient son client :

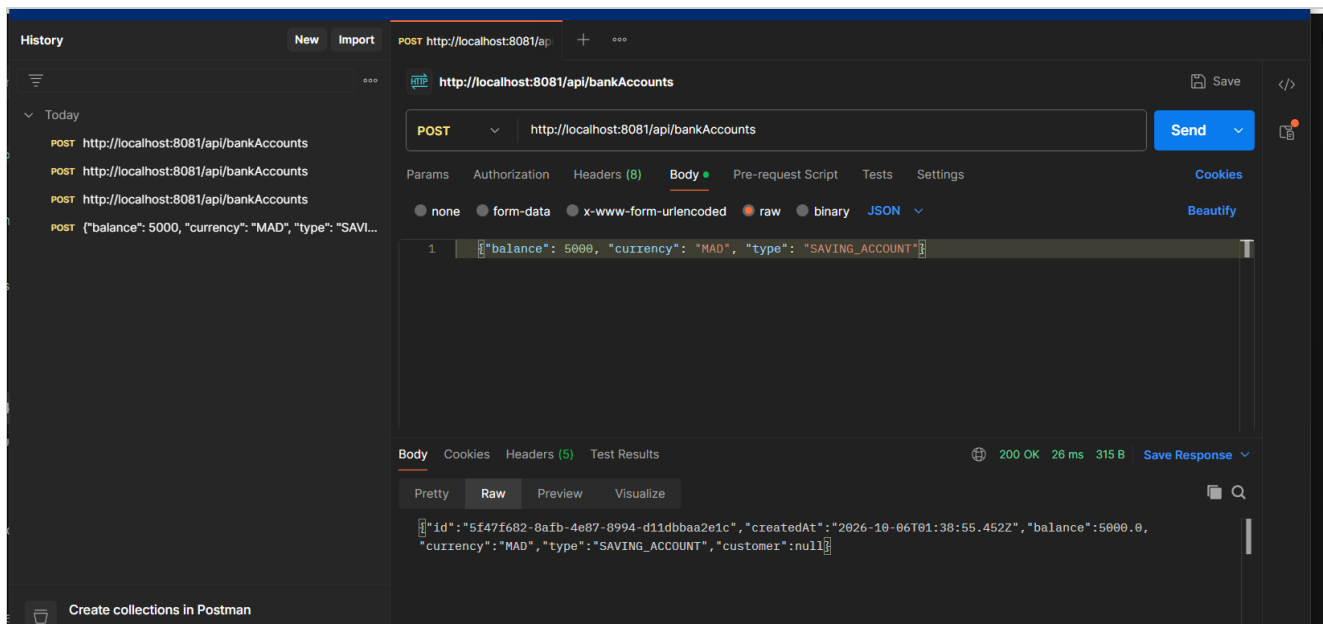
```
localhost:8081/api/bankAccounts
impression élégante
{"id":"7adc7fab-938b-4a20-9071-105d3d75e809","createdAt":"2026-10-06T01:31:29.623Z","balance":39448.06214250669,"currency":"MAD","type":"CURRENT_ACCOUNT","customer":{"id":1,"name":"Mohamed"}},{"id":"b598856a-dbef-1657-8bc6-1f5752a7ce95","createdAt":"2026-10-06T01:31:29.644Z","balance":140092.037320884934,"currency":"MAD","type":"CURRENT_ACCOUNT","customer":{"id":1,"name":"Mohamed"}},{"id":"3ef1b177-7e61-4ff9-8cab-7ed0de839eb","createdAt":"2026-10-06T01:31:29.646Z","balance":52860.79640778226,"currency":"MAD","type":"CURRENT_ACCOUNT","customer":{"id":1,"name":"Mohamed"}},{"id":"1fedc1be-555e-4e77-97f9-779519810a9","createdAt":"2026-10-06T01:31:29.648Z","balance":72968.54296496458,"currency":"MAD","type":"SAVING_ACCOUNT","customer":{"id":1,"name":"Mohamed"}},{"id":"79ecf23a-1cfl-4090-94fd-44f2b86daeb","createdAt":"2026-10-06T01:31:29.651Z","balance":74615.49556109487,"currency":"MAD","type":"SAVING_ACCOUNT","customer":{"id":1,"name":"Mohamed"}},{"id":"7031a04-e5f2-46f4-8bbf-9c3f7c3e9da","createdAt":"2026-10-06T01:31:29.655Z","balance":62274.776893789414,"currency":"MAD","type":"CURRENT_ACCOUNT","customer":{"id":1,"name":"Mohamed"}},{"id":"359665a8-04de-42cb-bf8a-238533174ba","createdAt":"2026-10-06T01:31:29.659Z","balance":98309.4822092622,"currency":"MAD","type":"SAVING_ACCOUNT","customer":{"id":1,"name":"Mohamed"}},{"id":"577fcb27-0c43-48ce-852b-3af4bf216d0","createdAt":"2026-10-06T01:31:29.662Z","balance":99300.84087661836,"currency":"MAD","type":"CURRENT_ACCOUNT","customer":{"id":1,"name":"Mohamed"}},{"id":"e14b4d3b-a437-45c4-b751-688623890e3","createdAt":"2026-10-06T01:31:29.664Z","balance":92109.16231259448,"currency":"MAD","type":"SAVING_ACCOUNT","customer":{"id":1,"name":"Mohamed"}},{"id":"fbc954ad-5451-462d-b84c-463cb266ed1","createdAt":"2026-10-06T01:31:29.668Z","balance":41479.08517877216,"currency":"MAD","type":"CURRENT_ACCOUNT","customer":{"id":1,"name":"Mohamed"}},{"id":"4d0469fc-ae9a-44f0-8ccd-34d69308fef","createdAt":"2026-10-06T01:31:29.673Z","balance":73168.06890806066,"currency":"MAD","type":"CURRENT_ACCOUNT","customer":{"id":2,"name":"Yassine"}},{"id":"b234d151-2502-4df9-bc13-f340ab7e087","createdAt":"2026-10-06T01:31:29.679Z","balance":34506.74446993385,"currency":"MAD","type":"SAVING_ACCOUNT","customer":{"id":2,"name":"Yassine"}},{"id":"7c5d39fc1-f17e-45a8-b783-4b5b9920fc31","createdAt":"2026-10-06T01:31:29.683Z","balance":60832.89060926137,"currency":"MAD","type":"CURRENT_ACCOUNT","customer":{"id":2,"name":"Yassine"}},{"id":"0f01445b-617e-4dae-a1f9-492ccfbfb78","createdAt":"2026-10-06T01:31:29.686Z","balance":75574.37234029082,"currency":"MAD","type":"SAVING_ACCOUNT","customer":{"id":2,"name":"Yassine"}},{"id":"08ec1dc8-d4c6-4542-ba20-9162e79eace","createdAt":"2026-10-06T01:31:29.690Z","balance":34744.32277721494,"currency":"MAD","type":"CURRENT_ACCOUNT","customer":{"id":2,"name":"Yassine"}},{"id":"905437ac-778f-49cf-abbe-9a98294287f6","createdAt":"2026-10-06T01:31:29.693Z","balance":57263.394391209935,"currency":"MAD","type":"SAVING_ACCOUNT","customer":{"id":2,"name":"Yassine"}},{"id":"6987c28e-7204-401e-b7ad-f8e196c9a6f","createdAt":"2026-10-06T01:31:29.697Z","balance":86505.64732057959,"currency":"MAD","type":"SAVING_ACCOUNT","customer":{"id":2,"name":"Yassine"}},{"id":"e8f2f9ce-2e51-470e-b357-0a63a66d44a","createdAt":"2026-10-06T01:31:29.701Z","balance":51998.3226777994,"currency":"MAD","type":"SAVING_ACCOUNT","customer":{"id":2,"name":"Yassine"}},{"id":"90da2f24-e082-4be2-ade7-66cb73d5722a","createdAt":"2026-10-06T01:31:29.705Z","balance":51770.34528982366,"currency":"MAD","type":"SAVING_ACCOUNT","customer":{"id":2,"name":"Yassine"}},{"id":"08c9517-52e7-4c3d-ba48-9f1ab1e9c648","createdAt":"2026-10-06T01:31:29.710Z","balance":45073.99936133888,"currency":"MAD","type":"SAVING_ACCOUNT","customer":{"id":2,"name":"Yassine"}},{"id":"0bf4376c-395-40b8-8a7f-770e7ba70ca2","createdAt":"2026-10-06T01:31:29.714Z","balance":81391.21848738356,"currency":"MAD","type":"CURRENT_ACCOUNT","customer":{"id":3,"name":"Hanae"}},{"id":"4f3274df-c43b-4f76-b371-f998df2d098","createdAt":"2026-10-06T01:31:29.719Z","balance":70225.30164160274,"currency":"MAD","type":"CURRENT_ACCOUNT","customer":{"id":3,"name":"Hanae"}},{"id":"1ad3f3d5-06f7-4e79-95b1-0ed12ba1d1","createdAt":"2026-10-06T01:31:29.722Z","balance":14076.876994829899,"currency":"MAD","type":"SAVING_ACCOUNT","customer":{"id":3,"name":"Hanae"}},{"id":"de87898e-f57c-4a1a-c5f9-1110101d450b","createdAt":"2026-10-06T01:31:29.726Z","balance":75466.9581355666,"currency":"MAD","type":"CURRENT_ACCOUNT","customer":{"id":3,"name":"Hanae"}},{"id":"dc3acf05-72bd-4877-aad4-a8d409906b99","createdAt":"2026-10-06T01:31:29.729Z","balance":76817.53795425009,"currency":"MAD","type":"CURRENT_ACCOUNT","customer":{"id":3,"name":"Hanae"}},{"id":"df856ade-d414-433b-9143-f52763a0e21a","createdAt":"2026-10-06T01:31:29.733Z","balance":58502.79072890222,"currency":"MAD","type":"CURRENT_ACCOUNT","customer":{"id":3,"name":"Hanae"}},{"id":"3ac00e65-0dab-4516-8625-805e826e2f66","createdAt":"2026-10-06T01:31:29.737Z","balance":87170.57533909888,"currency":"MAD","type":"CURRENT_ACCOUNT","customer":{"id":3,"name":"Hanae"}},{"id":"8b1f6749-259e-4556-93d7-45711062c400","createdAt":"2026-10-06T01:31:29.741Z","balance":20380.9534752456,"currency":"MAD","type":"SAVING_ACCOUNT","customer":{"id":3,"name":"Hanae"}},{"id":"86959140-c8eb-48be-adac-a77cba3f4d8b","createdAt":"2026-10-06T01:31:29.745Z","balance":59337.2749046418,"currency":"MAD","type":"SAVING_ACCOUNT","customer":{"id":3,"name":"Hanae"}},{"id":"516601f4-3025-43f6-b4e3-8bab58f5801","createdAt":"2026-10-06T01:31:29.748Z","balance":53332.77320474972,"currency":"MAD","type":"SAVING_ACCOUNT","customer":{"id":3,"name":"Hanae"}},{"id":"aa4f9cee-0a4b-4a58-aed5-8991489c871e","createdAt":"2026-10-06T01:31:29.751Z","balance":45815.783083964685,"currency":"MAD","type":"CURRENT_ACCOUNT","customer":{"id":4,"name":"Imane"}},{"id":"33ae001d-8b2b-4914-810e-8997e1240cc","createdAt":"2026-10-06T01:31:29.754Z","balance":73518.57913863557,"currency":"MAD","type":"CURRENT_ACCOUNT","customer":{"id":4,"name":"Imane"}},{"id":"2e497780-9b52-4094-b9f1-994c2365274b","createdAt":"2026-10-06T01:31:29.758Z","balance":79881.20556328174,"currency":"MAD","type":"SAVING_ACCOUNT","customer":{"id":4,"name":"Imane"}},{"id":"acacc0c0-9de4-40d8-96dd-0279a9f1e8f6","createdAt":"2026-10-06T01:31:29.761Z","balance":22074.22971255961,"currency":"MAD","type":"CURRENT_ACCOUNT","customer":{"id":4,"name":"Imane"}},{"id":"4594376d-404d-49d3-c5ef-52b5919e6d77","createdAt":"2026-10-06T01:31:29.763Z","balance":63222.39882795041,"currency":"MAD","type":"CURRENT_ACCOUNT","customer":{"id":4,"name":"Imane"}},{"id":"ba6b2b05-79da-4a56-8cbe-ae89943b4bf9","createdAt":"2026-10-06T01:31:29.767Z","balance":122727.04367682907,"currency":"MAD","type":"SAVING_ACCOUNT","customer":{"id":4,"name":"Imane"}},{"id":"a3fcbf9d-c3d2-4892-a3d1-9a448dd0ec520","createdAt":"2026-10-06T01:31:29.773Z","balance":26343.88312944772,"currency":"MAD","type":"SAVING_ACCOUNT","customer":{"id":4,"name":"Imane"}},{"id":"e3fd9144-77e4-446b-bc95-7cd1dc6e13c","createdAt":"2026-10-06T01:31:29.776Z","balance":64941.370345609655,"currency":"MAD","type":"CURRENT_ACCOUNT","customer":{"id":4,"name":"Imane"}},{"id":"15b067a8-d1da-4d9f-b0e9-0a340310095c","createdAt":"2026-10-06T01:31:29.779Z","balance":52412.56110638828,"currency":"MAD","type":"CURRENT_ACCOUNT","customer":{"id":4,"name":"Imane"}}
```

5. Test avec Postman

Ajout d'un compte — POST /api/bankAccounts avec le body JSON :

```
{ "balance": 5000, "currency": "MAD", "type": "SAVING_ACCOUNT" }
```

Le compte est créé avec un `id` et une date `createdAt` générés automatiquement (réponse 200 OK) :



6. Documentation Swagger

Dépendance ajoutée (version 3.x compatible Spring Boot 4) :

```
<dependency>
  <groupId>org.springdoc</groupId>
  <artifactId>springdoc-openapi-starter-webmvc-ui</artifactId>
  <version>3.0.0</version>
</dependency>
```

Remarque : avec Spring Boot 4, springdoc combiné à Spring Data REST nécessite aussi `spring-boot-starter-hateoas`, sinon l'application ne démarre pas (`HateoasProperties` not present).

Documentation accessible sur <http://localhost:8081/swagger-ui/index.html> :

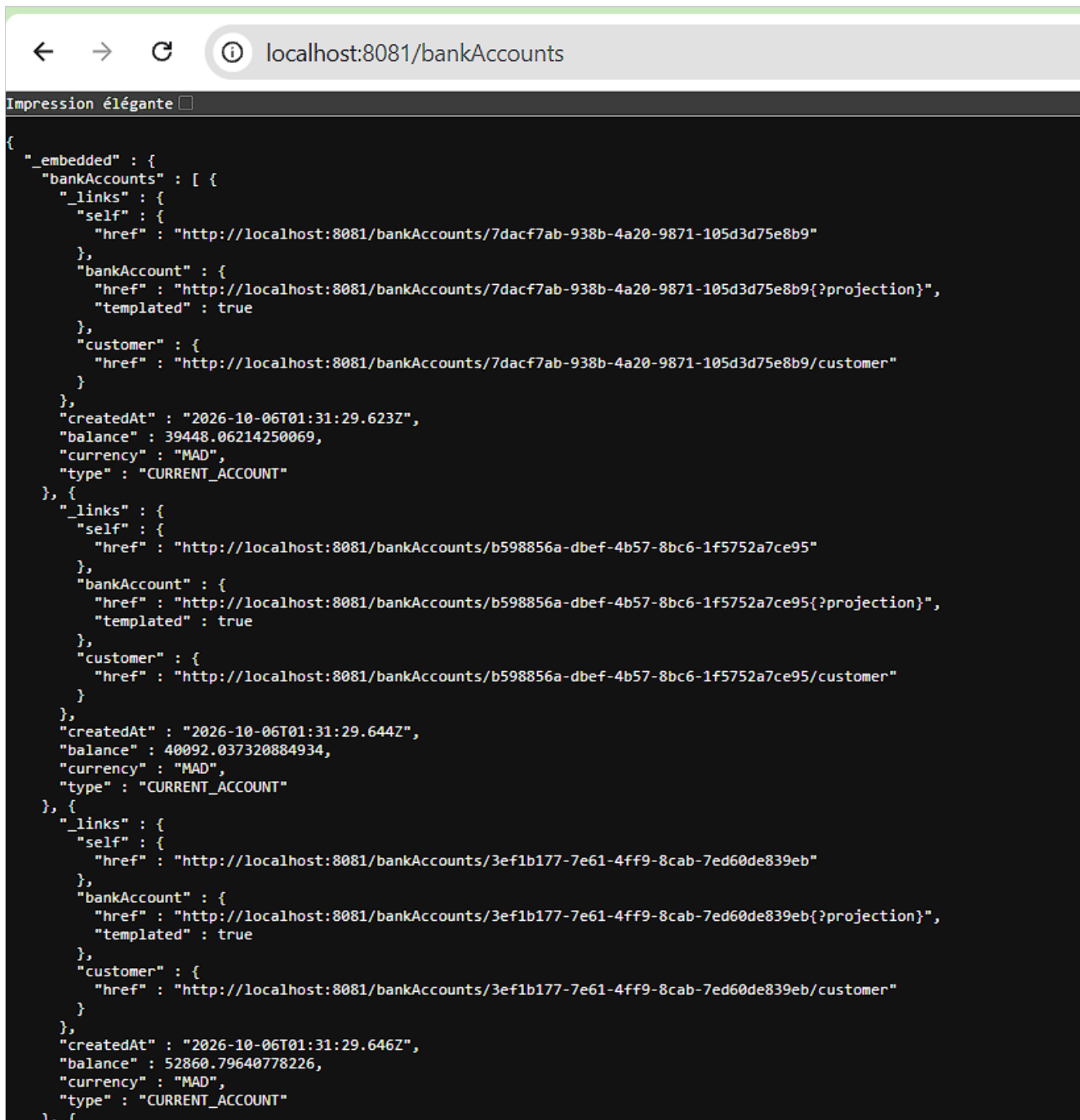
Servers	
http://localhost:8081 - Generated server url	
bank-account-entity-controller ^	
GET	/bankAccounts
POST	/bankAccounts
GET	/bankAccounts/{id}
PUT	/bankAccounts/{id}
DELETE	/bankAccounts/{id}
PATCH	/bankAccounts/{id}
bank-account-search-controller ^	
GET	/bankAccounts/search/byType
bank-account-property-reference-controller ^	
GET	/bankAccounts/{id}/customer
PUT	/bankAccounts/{id}/customer
DELETE	/bankAccounts/{id}/customer
PATCH	/bankAccounts/{id}/customer

7. Spring Data REST et projections

L'annotation `@RepositoryRestResource` expose automatiquement le repository en API REST (format HAL). Une méthode de recherche personnalisée est ajoutée :

```
@RestResource(path = "/byType")
List<BankAccount> findByType(@Param("t") AccountType type);
```

Endpoints générés (`http://localhost:8081/bankAccounts`) :



```
{
  "_embedded" : {
    "bankAccounts" : [ {
      "_links" : {
        "self" : {
          "href" : "http://localhost:8081/bankAccounts/7dacf7ab-938b-4a20-9871-105d3d75e8b9"
        },
        "bankAccount" : {
          "href" : "http://localhost:8081/bankAccounts/7dacf7ab-938b-4a20-9871-105d3d75e8b9{?projection}",
          "templated" : true
        },
        "customer" : {
          "href" : "http://localhost:8081/bankAccounts/7dacf7ab-938b-4a20-9871-105d3d75e8b9/customer"
        }
      },
      "createdAt" : "2026-10-06T01:31:29.623Z",
      "balance" : 39448.06214250069,
      "currency" : "MAD",
      "type" : "CURRENT_ACCOUNT"
    }, {
      "_links" : {
        "self" : {
          "href" : "http://localhost:8081/bankAccounts/b598856a-dbef-4b57-8bc6-1f5752a7ce95"
        },
        "bankAccount" : {
          "href" : "http://localhost:8081/bankAccounts/b598856a-dbef-4b57-8bc6-1f5752a7ce95{?projection}",
          "templated" : true
        },
        "customer" : {
          "href" : "http://localhost:8081/bankAccounts/b598856a-dbef-4b57-8bc6-1f5752a7ce95/customer"
        }
      },
      "createdAt" : "2026-10-06T01:31:29.644Z",
      "balance" : 40092.037320884934,
      "currency" : "MAD",
      "type" : "CURRENT_ACCOUNT"
    }, {
      "_links" : {
        "self" : {
          "href" : "http://localhost:8081/bankAccounts/3ef1b177-7e61-4ff9-8cab-7ed60de839eb"
        },
        "bankAccount" : {
          "href" : "http://localhost:8081/bankAccounts/3ef1b177-7e61-4ff9-8cab-7ed60de839eb{?projection}",
          "templated" : true
        },
        "customer" : {
          "href" : "http://localhost:8081/bankAccounts/3ef1b177-7e61-4ff9-8cab-7ed60de839eb/customer"
        }
      },
      "createdAt" : "2026-10-06T01:31:29.646Z",
      "balance" : 52860.79640778226,
      "currency" : "MAD",
      "type" : "CURRENT_ACCOUNT"
    }
  ]
}
```

Projection — l'interface `AccountProjection (@Projection(name = "p1"))` ne renvoie que `id`, `type` et `balance` (`http://localhost:8081/bankAccounts?projection=p1`) :

```
localhost:8081/bankAccounts?projection=p1

Impression élégante

{
  "_embedded" : {
    "bankAccounts" : [ {
      "_links" : {
        "self" : {
          "href" : "http://localhost:8081/bankAccounts/7dacf7ab-938b-4a20-9871-105d3d75e8b9"
        },
        "bankAccount" : {
          "href" : "http://localhost:8081/bankAccounts/7dacf7ab-938b-4a20-9871-105d3d75e8b9{?projection}",
          "templated" : true
        },
        "customer" : {
          "href" : "http://localhost:8081/bankAccounts/7dacf7ab-938b-4a20-9871-105d3d75e8b9/customer"
        }
      },
      "balance" : 39448.06214250069,
      "id" : "7dacf7ab-938b-4a20-9871-105d3d75e8b9",
      "type" : "CURRENT_ACCOUNT"
    }, {
      "_links" : {
        "self" : {
          "href" : "http://localhost:8081/bankAccounts/b598856a-dbef-4b57-8bc6-1f5752a7ce95"
        },
        "bankAccount" : {
          "href" : "http://localhost:8081/bankAccounts/b598856a-dbef-4b57-8bc6-1f5752a7ce95{?projection}",
          "templated" : true
        },
        "customer" : {
          "href" : "http://localhost:8081/bankAccounts/b598856a-dbef-4b57-8bc6-1f5752a7ce95/customer"
        }
      },
      "balance" : 40092.037320884934,
      "id" : "b598856a-dbef-4b57-8bc6-1f5752a7ce95",
      "type" : "CURRENT_ACCOUNT"
    }, {
      "_links" : {
        "self" : {
          "href" : "http://localhost:8081/bankAccounts/3ef1b177-7e61-4ff9-8cab-7ed60de839eb"
        },
        "bankAccount" : {
          "href" : "http://localhost:8081/bankAccounts/3ef1b177-7e61-4ff9-8cab-7ed60de839eb{?projection}",
          "templated" : true
        },
        "customer" : {
          "href" : "http://localhost:8081/bankAccounts/3ef1b177-7e61-4ff9-8cab-7ed60de839eb/customer"
        }
      },
      "balance" : 52860.79640778226,
      "id" : "3ef1b177-7e61-4ff9-8cab-7ed60de839eb",
      "type" : "CURRENT_ACCOUNT"
    }, {
      "_links" : {
        "self" : {
          "href" : "http://localhost:8081/bankAccounts/1fcd31bc-555c-4e77-97f9-2799510810a0"
        },
        "bankAccount" : {
          "href" : "http://localhost:8081/bankAccounts/1fcd31bc-555c-4e77-97f9-2799510810a0{?projection}",
          "templated" : true
        },
        "customer" : {
          "href" : "http://localhost:8081/bankAccounts/1fcd31bc-555c-4e77-97f9-2799510810a0/customer"
        }
      },
      "balance" : 52860.79640778226,
      "id" : "1fcd31bc-555c-4e77-97f9-2799510810a0",
      "type" : "CURRENT_ACCOUNT"
    }
  ]
}
```

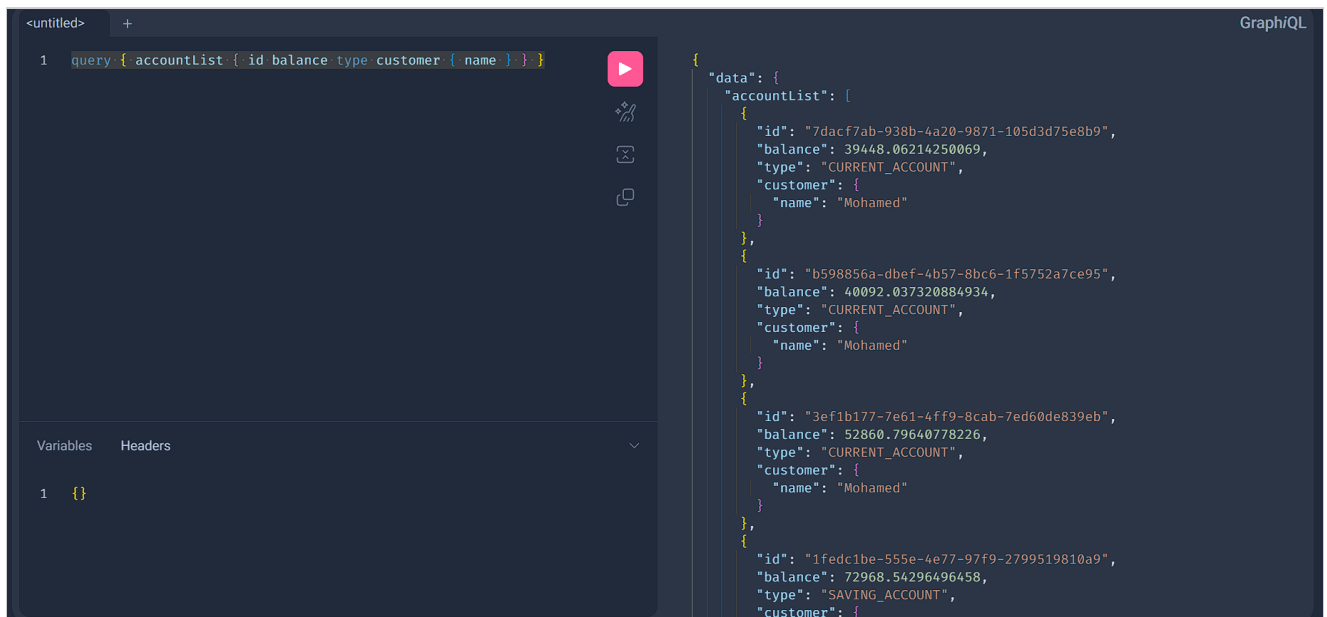
8. Service web GraphQL

Le schéma est défini dans `resources/graphql/schema.graphqls` (types `BankAccount`, `Customer`, input `BankAccountDTO`, `Query` et `Mutation`). Le contrôleur `BankAccountGraphQLController` utilise `@QueryMapping` et `@MutationMapping`. Une classe `CustomDataFetcherExceptionHandlerResolver` renvoie des messages d'erreur clairs.

Interface `GraphiQL` : `http://localhost:8081/graphiql?path=/graphql`

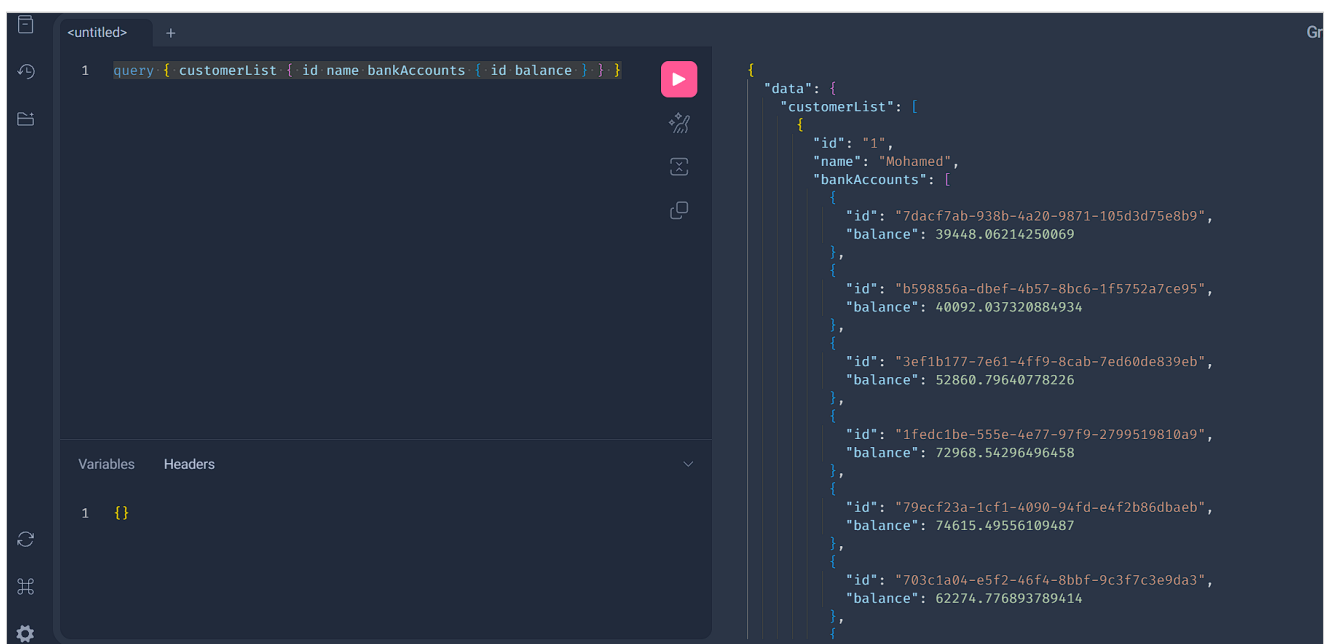
Liste des comptes avec leur client — on ne demande que les champs nécessaires :

```
query { accountList { id balance type customer { name } } }
```



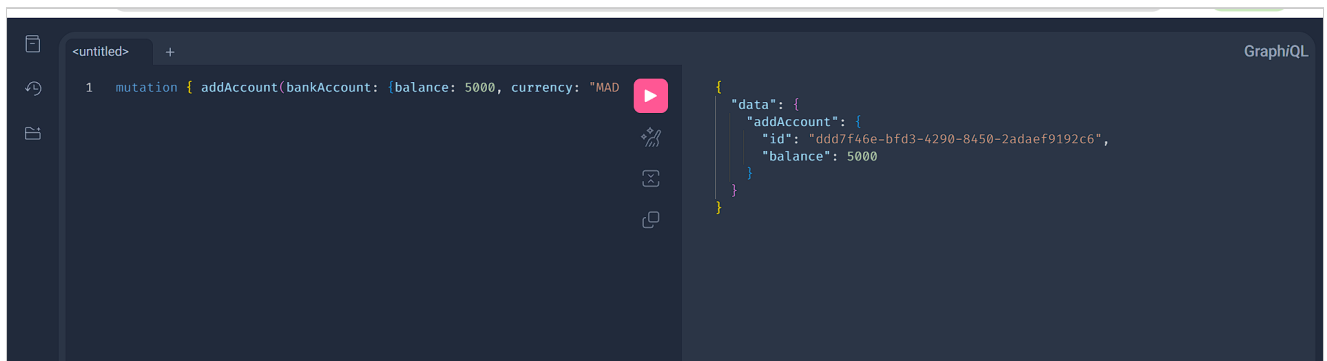
Liste des clients avec leurs comptes — contrairement à REST, GraphQL n'a pas de problème de récursivité :

```
query { customerList { id name bankAccounts { id balance } } }
```



Mutation — ajout d'un compte :

```
mutation { addAccount(bankAccount: {balance: 5000, currency: "MAD", type: "SAVING_ACCOUNT"})
{ id balance } }
```



Conclusion

Cette activité a permis de mettre en place un microservice Spring Boot complet et d'exposer les mêmes données de plusieurs façons : API REST manuelle avec une architecture en couches (entités, repositories, DTOs, mappers, services), API générée par Spring Data REST avec projections, documentation Swagger, et service GraphQL qui permet au client de choisir précisément les champs qu'il veut récupérer.